Chapter two

Ratios and Percentages

In our everyday life, we always compare the price, quantity etc. of one thing with those of another thing. That is, of more than one thing, we compare one thing to the other things in one was oranother. Here comes ratio from this comparison. Again, we compare one thing to the parts, to how many times, or how many parts in percentage of other things. Usually we know of all these through ratio and percentage. So, we should have clear conception about ratio and percentage. On the other hand, there is a relation between Percentage and Fraction. In this chapter, all mentioned topics have been presented.

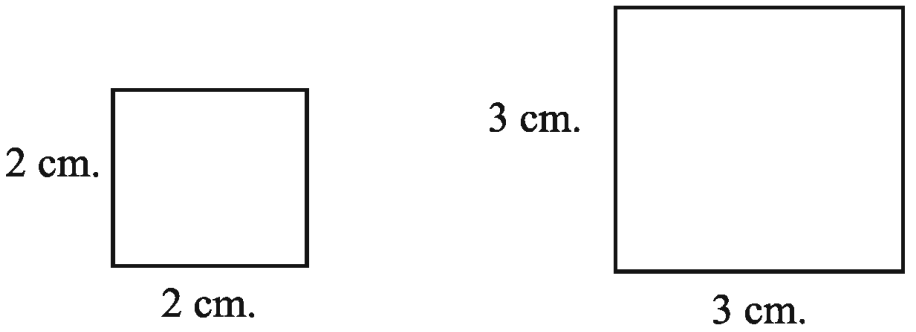
At the end of the chapter, the students will be able to

- explain what ratio is.
- solve the problems related to simple ratio.
- express percentage to fraction and fraction to percentage.
- express ratio to percentage and percentage to ratio.
- describe unitary method and the method of percentage accounting; and
- solve the arithmetic problems about time and work, time and food time and distance with the help of unitary method and the method of percentage accounting.

2·1 Ratio

In our everyday life we often compare two things of the same kind. For example, if the height of Nabil is 150 cm and that of his sister is 140 cm, we can say that the height of Nabil is (150 – 140) cm or 10 cm more than that of his sister. Thus we can make comparison by finding out the difference.

Again, if we want to compare the areas of two squares, then comparing by differences of area is not justified. Rather, we can make justified comparison of the areas of two squares by finding out how many times a square is greater or smaller than the other. We can make this comparison by dividing the area of a square by that of the other. This comparison with division is called ratio. ' : ' is the mathematical symbol for ratio.



In the above figures, we see that there are two squares of areas 4 sq. cm and 9 sq. cm. The ratio of areas of the squares is $\frac{4}{9} = 4 : 9$ or $\frac{9}{4} = 9 : 4$ depending on which comes first. We see that ratio is a fraction.
We observe the following examples :



(a) The rectangular region is divided into 7 equal parts, 2 parts are white and the rest 5 parts are black. The ratio of white and black parts is 2 : 5 and here 2 and 5 are called antecedent and subsequent respectively.

(b) Showkat's weight is 30 kg and that of his father is 60 kg. How many times is the weight of father greater than that of Showkat ?
Ratio of the weights of father and Showkat is

$$\frac{60 \text{ kg}}{30 \text{ kg}} = \frac{2}{1}$$ [Dividing the numerator & denominator by 30]

$$= 2 : 1$$

Here, the weight of father is twice the weight of Showkat.

(c) In a class, there are 50 boys and 40 girls
Here, the ratio of the number of boys and girls is

$\frac{50}{40} = \frac{5}{4}$ [dividing the numerator & denominator by 10]

$= 5 : 4$

Is it possible to compare the age of a child to the weight of a child ? It can never be done. To compare, two objects should be of the same kind. Further, let us suppose that the age of a baby is 6 years and that of another baby is 9 years 6 months. Though the ages are of the same kind, they can not be compared directly. The two objects should be of the same unit. In this case, the age of two babies should be expressed either in years or in months.

Here, 6 years = (6×12) months = 72 months ($\because$ 1 year = 12 months)
9 years 6 months = $(9 \times 12 + 6)$ months = 114 months.
So, the ratio of age of two babies is 72 : 114 or 12 : 19.

Let us consider that the age of brother is 3 years and that of sister is 6 months. We shall have to find the ratio of their ages.

Brother's age = 3 years
= 3×12 months ($\because$ 1 year = 12 months)
= 36 months

Sister s age = 6 months

$\therefore$ Ratio of their ages

$= \frac{36 \text{ months}}{6 \text{ months}}$ or $\frac{36}{6}$ or $\frac{6}{1}$ [Dividing numerator and denominator by 6]

$= 6 : 1$

Observe: It is to be noted that the different units can not be compared with each other. To compare, the units are to be of the same kind. Such as, the year has been converted into months in the above example.

Of two quantities of the same kind one is how many times or parts compared to the other can be expressed by fraction. This fraction is called ratio of two quantities. Since the two quantities are of the same kind, the ratio has no unit.

Activity :

1. Determine the ratio of numbers of your khatas and books.
2. Determine the ratio of breadth and wide of your math book.
3. Determine the ratio of length and breadth of the table of your class.

2·2 Different Ratios

Equivalent Ratios

The value of a ratio remains unchanged when its antecedent and subsequent are multiplied or divided by any number except 0. Such ratios are called equivalent ratio.

For example, $2 : 5 = \frac{2}{5} = \frac{2 \times 2}{5 \times 2} = \frac{4}{10} = 4 : 10$

$\therefore$ 2 : 5 and 4 : 10 are equivalent ratios.

There are an infinite number of equivalent ratios of any ratio.
Such as , 2 : 3, 4 : 6, 6 : 9 and 8 : 12 are all equivalent ratios.
Again, in 1 : 2 = 5 :□, if we put 10 in the blanks then the ratio will be an equivalent ratio.

Observe :

- The ratio can be simplified if the terms of the ratio are divided by H.C.F. (Highest Common Factor).
- Each part of the ratio can be found if they are divided by the sum of the antecedent and subsequent of the ratio.

Example 1 : The ratio of age of Jesmin and Abida is 3:2. The present age ratio of Abida and Anika is 5:1. The present age of Anika is 3 years 6 months.

(a) Express the first ratio in the stem into percentage.
(b) What will be Abida's age after 5 years?
(c) What is the percentage of Jesmin's present age to Anika's present age?

Solution:

(a) The first ratio of the stem = 3.2

$$= \frac{3}{2}$$

$$= \frac{3 \times 100}{2 \times 100}$$

$$= \left(\frac{3 \times 100}{2}\right)\%$$

$$= 150\%$$

(b) Present age of Abida : present age of Anika =5:1. It means Abida's present age is 5 times more than Anika's.

Anika's present age = 3 years 6 months

$= (3 \times 12+6)$ months; [$\because$ 1 year = 12 months]

$= (36+6)$ months

$= 42$ months

So, Abida's present age $= (42 \times 5)$ months

$= 210$ months

$= \frac{210}{12}$ years [$\because$ 12 months = 1 year]

$= \frac{\cancel{210}^{35}}{\cancel{12}_{2}}$ years

$= \frac{35}{2}$ years,

$= 17\frac{1}{2}$ years.

$\therefore$ after 5 years Abida's age will be

$= (17\frac{1}{2}+5)$ years

$= 22\frac{1}{2}$ years

(c) The ratio of ages of Jesmin and Abida is 3:2

It means Jesmin's present age is $\frac{3}{2}$ times more than Abida's present age.

From 'b' Abida's present age = $17\frac{1}{2}$ years

$\therefore$ Jesmin's present age $= 17\frac{1}{2} \times \frac{3}{2}$

$= \left(\frac{35}{2} \times \frac{3}{2}\right)$

$= \frac{105}{4}$

$= 26\frac{1}{4}$ years

Anika's present age = 3 years 6 months

$= 3\frac{6}{12}$ years [$\because$ 1 year = 12 months]

$$= 3\frac{1}{2} \text{ year}$$

$$= \frac{7}{2} \text{ year}$$

$\therefore$ Anika's present age is Jesmin's age's

$$= \left(\frac{7}{2} \div 26\frac{1}{4}\right) \text{portion}$$

$$= \left(\frac{\cancel{7}}{\cancel{2}} \times \frac{\cancel{4}^{2}}{\cancel{105}_{15}}\right) \text{ portion}$$

$$= \frac{2}{15} \text{ portion}$$

$$= \left(\frac{\cancel{2}}{\cancel{15}_{3}} \times \cancel{100}^{20}\right)\%$$

$$= \frac{40}{3}\%,$$

$$= 13\frac{1}{3}\%$$

So, Anika's present age is $13\frac{1}{3}\%$ of Jesmin's present age.

Example 2. Divide Tk. 500 between two labours in the ratio 2 : 3.

Solution : The antecedent of the ratio is 2 and the subsequent of that is 3. The sum of two quantities is $2 + 3 = 5$.

$\therefore$ 1st labour will get $\frac{2}{5}$ times of Tk. 500 $= \text{Tk. } 500 \times \frac{2}{5} = \text{Tk. } 200$

2nd labour will get $\frac{3}{5}$ times of Tk. 500 $= \text{Tk. } 500 \times \frac{3}{5} = \text{Tk. } 300$

Activity :

1. If the age of Mamun is 4 years and that of his sister is 6 months, determine the ratio of their age.
2. The height of Sajal and Sujan are 1m 75 cm and 1m 50 cm respectively. Determine the ratio of their height.

Simple Ratio

The ratio of two quantities is called **simple ratio**.

The first quantity of simple ratio is called antecedent and the second quantity is called subsequent.

Such as, 3 : 5 is a simple ratio. Here, 3 is antecedent and 5 is subsequent.

Ratio of less inequality

If the antecedent is smaller than the subsequent of a simple ratio, then the ratio is called the **ratio of less inequality**. For example 3 : 5 and 4 : 7 both are the ratios of less inequality.

In a school, the average age of the students of class III is 8 years and that of the students of class V is 10.
Here, the ratio of the average ages of the students of class III and class V is 8 : 10 or 4 : 5.

In this ratio, the antecedent being smaller than the subsequent, it is called the **ratio of less inequality**.

Ratio of greater inequality

If the antecedent is greater than the subsequent of a simple ratio, it is called the **ratio of greater inequality**. Such as 5 : 3, 7 : 4 and 6 : 5 all are the ratios of greater inequality.

Sadia bought a packet of biscuit by Tk. 32 and a cone ice-cream by Tk. 25. Here, the ratio of prices of biscuit and ice-cream is 32 : 25. The antecedent 32 of this ratio being greater than the subsequent 25, it is the **ratio of greater inequality.**

Unit Ratio

The ratio in which the antecedent and the subsequent are equal is called **unit ratio**.

For example, Arif bought a ballpen by Tk. 15 and a khata by Tk. 15. Here, the prices of ballpen and khata are the same and the ratio of prices is 15 : 15 or 1 : 1. So, it is unit ratio.

Inverse Ratio

The ratio formed by interchanging the antecedent and subsequent of a simple ratio is called **inverse ratio.**

For example, 5 : 13 is the inverse ratio of a simple ratio 13 : 5.

Mixed or Compound Ratio

The ratio obtained by whose antecedent and subsequent are formed by the products of antecedents and subsequent of more than one simple ratio is called **mixed or compound ratio.**

Such as, the mixed ratio of simple ratios 2 : 3 and 5 : 7 is
$(2 \times 5) : (3 \times 7) = 10 : 21$.

Example 3. Determine the compound ratio of the given simple ratios 5 : 7, 4 : 9, 3 : 2.

Solution : The product of the antecedents of the given three ratios is
$5 \times 4 \times 3 = 60$.
and the product of subsequent is $7 \times 9 \times 2 = 126$
$\therefore$ Required compound ratio is 60 : 126 or 10 : 21.

Activity :
(1) Express as the inverse ratio of the simple ratio 4 : 9
(2) Determine the antecedents and subsequents of the following ratios
(a) 4 : 11 (b) 7 : 5 (c) 19 : 21.
(3) Which one of the following is unit ratio ?
(a) 2 : 5 (b) 5 : 7 (c) 11 : 11
(4) Divide the followings into the ratio of less inequality and the ratio of greater inequality :
(a) 13 : 19 (b) 7 : 12 (c) 25 : 13 (d) 27 : 7
(5) Determine the mixed ratio of the following ratios 2 : 3 and 3 : 4.

Example 4. The sum of two numbers is 360. If the ratio of the numbers is 4 : 5, find the two numbers.
Solution : The ratio of numbers is 4 : 5.
The sum of antecedent and subsequent is $4 + 5 = 9$.

$$\text{The first number} = \frac{4}{9} \text{ times of } 360.$$

$$= \overset{40}{\cancel{360}} \times \frac{4}{\cancel{9}} = 160.$$

$$\text{The second number} = \frac{5}{9} \text{ times of } 360.$$

$$= \overset{40}{\cancel{360}} \times \frac{5}{\cancel{9}} = 200.$$

Required two numbers are 160 and 200.

Example 5. In a mixture of 40 kg, the ratio of sand and cement is 4 : 1. Determine the quantity of sand and cement in the mixture.

Solution : The quantity of mixture is 40 kg. The ratio of sand and cement is 4 : 1. Here, the sum of antecedent and subsequent = 4 + 1 = 5.

$\therefore$ The quantity of sand = $\frac{4}{5}$ times of 40 kg

$= \cancel{40}^{8} \times \frac{4}{\cancel{5}}$ kg $= 32$ kg

The quantity of cement = $\frac{1}{5}$ times of 40 kg

$= \cancel{40}^{8}$ kg $\times \frac{1}{\cancel{5}}$ kg $= 8$ kg

Example 6. The ratio of numbers of boys and girls of a school is 5 : 7. If the number of girls is 350, what is the number of boys ?

Solution : Number of boys : Number of girls = 5 : 7

i.e. the number of boys is $\frac{5}{7}$ times of number of girls.

Given that the number of girls is 350

$\therefore$ Number of boys = $\cancel{350}^{50} \times \frac{5}{\cancel{7}} = 250$

Required number of boys is 250.

Exercise 2·1

1. Express the following pairs of number as the ratio of the first and second quantities :

   (a) 25 and 35 (b) $7\frac{1}{3}$ and $9\frac{2}{5}$ (c) 1 year 2 months and 7 months

   (d) 7 kg and 2 kg 300 gram (d) Tk. 2 and 40 paisa

2. Simplify the following ratios.

   (a) 9 : 12 (b) 15 : 21 (c) 45 : 36 (d) 65 : 26

3. Fill in the gaps of the following equivalent ratios :

(a) 2 : 3 = 8 : ☐ (b) 5 : 6 = ☐ : 36 (c) 7 : ☐ = 42 :54

(d) ☐ : 9 = 63 : 81

4. The ratio of length and breadth of a hall room is 2 : 5. Putting the possible values of length and breadth in the gaps, complete the following table.

Breadth of hall room (m.)	10		40		160
Length of hall room (m.)	25	50		200	

5. Identify the equivalent ratios of the followings :

12 : 18; 6 : 18; 15 : 10; 3 : 2; 6 : 9; 2 : 3 ; 1 : 3; 2 : 6 ; 12 : 8

6. Express the following simple ratios into the mixed ratio :
(a) 3 : 5, 5 : 7 and 7 : 9 (b) 5 : 3, 7 : 5 and 9 : 7

7. Express the ratio 9 : 16 as inverse ratio.

8. Which one of the following ratios are unit ratios :
(a) 16 : 13 (b) 13 : 17 (c) 21 : 21

9. Divide Tk. 550 in the ratios of 5 : 6 and 4 : 7.

10. The ratio of the ages of father and son is 14 : 3. If the age of father is 56 years, what is the age of son ?

11. The sum of two numbers is 630. Their ratio is 10 : 11. Determine two numbers.

12. The ratio of the prices of two books is 5 : 7. If the price of the second book is 84, what is the price of the first book ?

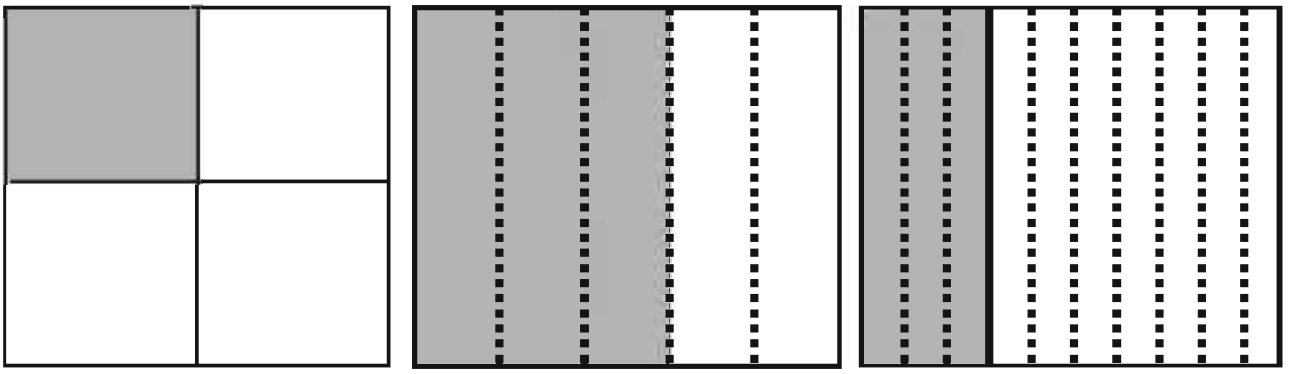
13. In the golden ornaments of 20 grams of 18 carate the ratio of gold and alloy is 3 : 1. Determine the quantity of gold and alloy of those ornaments.

14. The ratio of time of coming and going to school from houses of two friends is 2 : 3. If the distance of school from the house of the first friend is 5 km., then what is the distance of school from the house of the second friend ?

15. The ratio of milk and sugar in frumenty is 7 : 2. In the frumenty the quantity of sugar is 4 kg. what is the quantity of milk ?

16. The ratio of the prices of two computers is 5 : 6. If the price of the first computer is Tk. 25,000/-, what is the price of the second computer ? In case of price hike if the price of the first computer is increased by Tk. 5,000/- then what types of ratio of their prices will be?

2.3 Relation between Ratio and Percentage



1 : 4 a 3 : 5 b 3 : 10 c

Of the above figures, $\frac{1}{4}$ parts of a, $\frac{3}{5}$ parts of b, and $\frac{3}{10}$ parts of c have been shaded.

Here, we see that,

in figure (a), the ratio of the shaded part and the whole part is

$$1 : 4 = \frac{1}{4} = \frac{1 \times 25}{4 \times 25} = \frac{25}{100} = 25\%,$$

in figure (b), the ratio of the shaded part and the whole part is

$$3 : 5 = \frac{3}{5} = \frac{3 \times 20}{5 \times 20} = \frac{60}{100} = 60\%,$$

in figure (c), the ratio of the shaded part and the whole part is

$$3 : 10 = \frac{3}{10} = \frac{3 \times 10}{10 \times 10} = \frac{30}{100} = 30\%,$$

That is, 25%, 60%, 30% of the figures a, b, c are shaded respectively. It is seen that both percentage and ratio are fractions. But, in case of percentage, the denominator of the fraction is 100. In case of ratio numerator and the denominator may be of any natural number. If necessary, the percentage can be expressed into ratio and the ratio can be expressed into percentage. For example, the ratio of Tk. 7 and Tk. 10

$$= \frac{\text{Tk.}7}{\text{Tk.}10} = \frac{7}{10} = \frac{70}{100} = 70\%$$

Here, Tk. 7 is $\frac{7}{10}$ part of Tk.10 or $\frac{7}{10}$ times of Tk.10, which is equal to 70%.

On the other hand, percentage of 3 or 3% is $\frac{3}{100}$ or 3 : 100.

That is, a ratio can be expressed into percentage.

Activity :

1. Express the ratios 3 : 4 and 5 : 7 to percentage.
2. Express 5% and 12% to ratio.

Example 7. Express into ratios and decimals.

(a) 15% (b) 32% (c) 25% (d) 55% (e) $8\frac{1}{10}\%$

Solution : (a) $15\% = \frac{15}{100} = \frac{3}{20} = 3 : 20 = \cdot 15$

$\therefore 15\% = 3 : 20 = \cdot 15$

(b) $32\% = \frac{\overset{8}{\cancel{32}}}{\underset{25}{\cancel{100}}} = \frac{8}{25} = 8 : 25 = \cdot 32$

$\therefore 32\% = 8 : 25 = \cdot 32$

(c) $25\% = \frac{25}{100} = \frac{1}{4} = 1 : 4 = \cdot 25$

$\therefore 25\% = 1 : 4 = \cdot 25$

(d) $55\% = \frac{55}{100} = \frac{11}{20} = 11 : 20 = \cdot 55$

$\therefore\ 55\% = 11 : 20 = \cdot 55$

(e) $8\frac{1}{10}\% = \frac{81}{10}\% = \frac{81}{10} \times \frac{1}{100} = \frac{81}{1000} = 81 : 1000 = 0.081$

$\therefore\ 8\frac{1}{10}\% = 81 : 1000 = \cdot 081$

Example 8. Express the following fractions into percentage.

(a) $\frac{1}{4}$ (b) $\frac{3}{20}$ (c) $\frac{7}{15}$ (d) $\frac{4}{25}$ (e) $\frac{6}{13}$

Solution : (a) $\frac{1}{4} = \frac{1 \times 100}{4 \times 100} = \frac{25}{100} = 25\%$

(b) $\frac{3}{20} = \frac{3 \times 100}{20 \times 100} = \frac{15}{100} = 15\%$

(c) $\frac{7}{15} = \frac{7 \times 100}{15 \times 100} = \frac{140}{3} \times \frac{1}{100} = \frac{140}{3}\% = 46\frac{2}{3}\%$

(d) $\frac{4}{25} = \frac{4 \times 100}{25 \times 100} = \frac{16}{100} = 16\%$

(e) $\frac{3}{13} = \frac{3 \times 100}{13 \times 100} = \frac{300}{13} \times \frac{1}{100} = \frac{300}{13}\% = 23\frac{1}{13}\%$

Example 9. One quantity is 50% of another quantity. Determine the ratio of two quantities.

Solution : $50\% = \frac{50}{100}$ that is, if one quantity is 50, then the other will be 100.

$\therefore$ Ratio of 50 and 100 is $50 : 100 = 1 : 2$.

$\therefore$ Required ratio of two quantities $= 1 : 2$.

Example 10. The sum of two quantities is 240. If their ratio is 1 : 3, then find them. What is the percentage of the first quantity to the second ?

Solution : the Sum of two quantities = 240

Their ratio = 1 : 3

The sum of two quantities of the ratio = 1 + 3 = 4

$\therefore$ 1st quantity = $\frac{1}{4}$ times of 240

$= \overset{60}{\cancel{240}} \times \frac{1}{\cancel{4}} = 60$

$\therefore$ 2nd quantity = $\frac{3}{4}$ times of 240

$= \overset{60}{\cancel{240}} \times \frac{3}{\cancel{4}} = 180$

Again, ratio of two quantities is 1 : 3

$\therefore$ 1st quantity = $\frac{1}{3}$ of 2nd quantity = $\frac{1\times100}{3\times100} = \frac{100}{3}\% = 33\frac{1}{3}\%$

$\therefore$ 1st quantity is $33\frac{1}{3}\%$ of the second.

Example 11. Monira has obtained 80% marks in the annual examinations. If the total marks of the examinations are 800, how many marks has Monira obtained ?

Solution : The marks obtained by Monira = 80% of 800 = $\frac{80}{100} \times 800 = 640$

$\therefore$ the marks obtained by Monira 640

Example 12. 180 fazley mangoes were bought from a fruits shop. After two days 9 mangoes rotted. How many mangoes are quite good ?

Solution : The total number of mangoes which were bought = 180.
the number of rotten mangoes = 9.

$\therefore$ the number of good mangoes = 180 – 9 = 171.

The ratio of good mangoes and total mangoes is $\frac{171}{180} = \frac{19}{20}$

$\therefore$ The percentage of good mangoes is $\frac{19\times100}{20} = 95.$

Exercise 2·2

1. Express as percentage :
   (a) $\frac{3}{4}$ (b) $\frac{7}{15}$ (c) $\frac{4}{5}$ (d) $2\frac{6}{25}$ (e) 0.25
   (f) .65 (g) 2.50 (h) 3 : 10 (i) 12 : 25
2. Express as fractions and decimals :
   (a) 45% (b) $12\frac{1}{2}\%$ (c) $37\frac{1}{2}\%$ (d) $11\frac{1}{4}\%$
3. Calculate the followings :
   (a) 5% fo 125 ? (b) 9% of 225 ?
   (c) 6% of 6 kg rice ? (d) 40% of 200 cm ?
4. (a) What is the percentage of Tk. 20 of Tk. 80 ?
   (b) What is the percentage of Tk. 75 of Tk. 120 ?
5. In a school, there are 500 students. If the girls are 40%, then determine the number of boys ?
6. David has obtained 600 marks out of 900 in the terminal examinations. What is the percentage of marks he has obtained? Determine the ratio of the total and the obtained marks.
7. Musanna bought a Bengali essay book from book stall at Tk. 84. But the cover price of the book was Tk.120. How much commission in percentage has he obtained ?
8. The monthly income of a service holder is Tk. 15,000/-. His monthly expenditure is Tk. 9000. What is the percentage of the expenditure of the income ?
9. The monthly school fees of Shoeb are Tk. 200. His mother gives him Tk. 20 as tiffin expenses. What is the percentage of expenditure for his daily tiffin to the monthly school fees ?
10. In a school, there are 800 students. At the beginning of the year, if 5% new students get admitted, then what is the number of students at present in that school ?
11. In a class, 5% students were absent out of 200 students. How many students were present ?
12. Zahed, buying a book at 10% commission, gave Tk. 180 to the shopkeeper. What is the actual price of the book ?